UniRAP: Towards Unified Part-level Physical Affordance Reasoning and Actionable Perception

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Abstract

Vision-language perception has achieved impressive progress in aligning natural language with visual observations, yet grounding high-level semantics into part-level physical interaction remains challenging. To address this gap, we propose UniRAP, a unified model for inferring part-level physical affordances and mapping language instructions to actionable geometric representations. UniRAP formulates this problem as a conditional multimodal generation task, integrating visual inputs, textual instructions, and optional prompts into a shared spatiotemporal representation space through a unified interface token mechanism. To predict executable contact geometry, we introduce a Unified Affordance Decoder (UAD), which jointly performs object detection, part-level affordance segmentation, and 4-DoF interaction pose estimation by leveraging intermediate segmentation features. In addition, we propose a curriculum-based transfer training strategy that progressively adapts the model from general visual parsing to interaction-aware perception, improving data efficiency under limited textual annotations. Experiments show that UniRAP achieves state-of-the-art performance on referring expression segmentation, affordance grounding, and interaction pose estimation, while maintaining strong spatiotemporal consistency in dynamic video scenarios. These results demonstrate the effectiveness of UniRAP as a unified perception framework for language-guided physical manipulation.

1 Introduction

Vision-Language Perception (VLP) aims to establish semantic correspondences between natural language and visual observations, enabling models to interpret and localize relevant elements in a scene conditioned on linguistic instructions [1, 2]. With the advancement of Vision-Language Models (VLMs), these systems have demonstrated strong capabilities in cross-modal alignment and semantic understanding in open-world settings, achieving notable success in tasks such as visual question answering [3–7]. Building upon these foundations, subsequent works have extended such models to incorporate spatial perception, enabling them to generate coarse visual grounding from text, such as bounding boxes [8–13] or segmentation masks [14–19]. However, these capabilities remain largely confined to semantic understanding and alignment, lacking the fine-grained, actionable modeling required for physical interaction.

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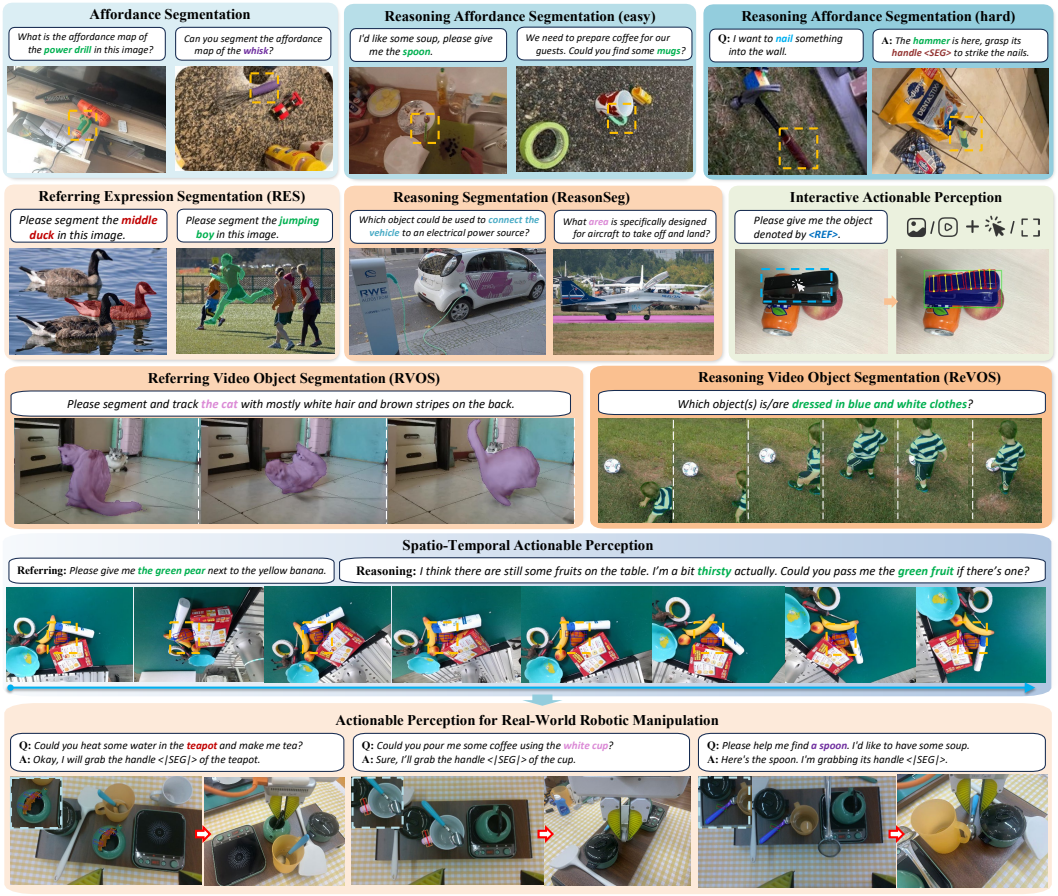


Figure 1: **UniRAP** transfers large-scale generic segmentation pretraining to affordance segmentation and spatiotemporally actionable perception, improving language grounding for task-relevant regions and dynamics. It enhances referring segmentation, affordance grounding, and interaction modeling while maintaining strong spatio-temporal consistency.

In embodied intelligence scenarios, this limitation becomes particularly critical: systems must not only understand `what` and `where`, but also further reason about `how to contact` [20, 21]. This requires models to possess **part-level actionable perception**, i.e., the ability to model interaction-relevant functional properties (affordances) [22, 23] and their corresponding operability, grounded in pixel-level spatial alignment and part-level structural understanding [24–27]. Without such capability, vision-language models struggle to translate language instructions into executable action representations, thereby limiting their applicability in real-world physical environments [28–30].

Despite notable progress, existing approaches still exhibit systematic limitations in advancing toward actionable perception. On one hand, cross-modal methods [31–33, 27, 34] based on CLIP/DINO-style visual representation alignment rely on static embedding spaces, which are inherently limited in modeling complex linguistic structures and compositional relationships. On the other hand, although VLM-based approaches demonstrate strong multi-step reasoning capabilities, their modeling granularity remains predominantly object-centric. Even when augmented with intermediate representations such as keypoints [35–41], detection [42, 13, 12, 43], or segmentation [44–46, 14, 15], these methods still fail to explicitly capture part-level functional affordances and contact regions, all of which are essential for reasoning about real-world interaction feasibility. Furthermore, RealVLG-R1 [47] attempts to predict 4-DoF grasp poses via structured language generation. However, this approach relies on an autoregressive generation mechanism, which is prone to limited precision and instability, making it inadequate for pixel-level actionable tasks.

To address the aforementioned challenges, we propose **UniRAP**, a **Uni**fied vision-language model for **R**easoning-Aware **A**ctionable **P**erception. As shown in Fig. 1, UniRAP supports both textual and visual prompts (e.g., points and bounding boxes) and is capable of producing multi-granularity

affordance perception outputs over continuous data modalities, including images and videos. These outputs include object detection bounding boxes, part-level affordance segmentation masks, and interaction-oriented structured representations. Notably, UniRAP adopts a 4-DoF grasp pose as an explicit representation for actionable perception, which establishes a principled correspondence between high-level linguistic semantics and physical interaction.

To mitigate data imbalance and cross-task interference arising from heterogeneous perceptual granularity across tasks, a unified prompting and decoding paradigm is designed. Specifically, a structured `<REF>` token is introduced to integrate explicit visual prompts and implicit textual references into a shared `<SEG>` interface, thereby enabling consistent modeling of diverse localization signals. Furthermore, a Unified Affordance Decoder (UAD) is proposed to facilitate cross-task representation sharing through the reuse of intermediate features, while preserving general segmentation capability to the greatest extent possible. Finally, a curriculum-based transfer training strategy is adopted, allowing the model to progressively acquire stable and consistent actionable perception capabilities without compromising its original reasoning performance.

The main contributions of this work are summarized as follows:

1. We propose UniRAP, a unified model that maps multimodal prompts to fine-grained physical contact geometry. It produces consistent multi-granularity affordance representations across both image and video domains.
2. A unified prompting and decoding pipeline is designed, together with a transfer-based training strategy, to alleviate task imbalance in cross-task learning and enable synergistic optimization across heterogeneous perception tasks.
3. Extensive experimental results demonstrate that UniRAP achieves a favorable trade-off between accuracy and stability across a range of tasks, including general segmentation, affordance perception, and part-level interaction grounding.

2 Related Work

Non-LLM Vision-Language Perception. Inspired by vision–language pretraining models such as CLIP [48, 49] and DINO [50, 51], early VLP research primarily addresses coarse-grained tasks, including open-vocabulary object detection [52–64] and semantic segmentation [65, 26, 66–71]. These methods rely on region–text alignment to associate visual and linguistic representations, enabling generalization beyond fixed category spaces. Subsequent studies extend VLP to affordance-aware perception, shifting from object recognition to modeling functional properties, typically formulated as part-level segmentation [22, 72–79] or region-level localization [80–83]. More recently, this paradigm further expands to actionable perception, such as contact point prediction [84–87] and 4-DoF grasp pose estimation [31, 32, 88–90], aiming to bridge the gap between visual perception and physical manipulation. However, due to limited textual understanding in pretrained models, these methods remain restricted to simple category-level language and struggle with compositional and reasoning-intensive instructions, which limits their applicability in open-world embodied scenarios.

LLM-based Vision-Language Perception. In recent years, large-scale VLMs have demonstrated strong semantic understanding capabilities in tasks such as visual question answering [3, 5, 7, 91, 8, 4, 92]. However, due to their reliance on global or region-level representations, they remain limited in pixel-level visual perception. To address this limitation, some approaches [93, 94, 12, 95, 13, 8] enforce the generation of structured language through SFT [96, 97] or RLVR [98, 99], enabling coarse-grained object localization. For dense segmentation tasks, VisionLLM [100] adopts a text-based polygon output paradigm, which is highly dependent on large-scale training data and still suffers from limited accuracy. To overcome these issues, LISA [14] introduces the "Embedding as Mask" paradigm, which extracts implicit features from the language model via a special token and injects them into SAM [24], enabling stable pixel-level predictions. Building on this line, subsequent studies further advance decoder design [101–104], scale up training data [105], and extend capabilities to video settings [106–108, 16]. UniPixel [15] further proposes the PixelQA task, unifying segmentation and region-referred visual question answering. In the context of affordance perception, existing methods largely remain limited to embodied question answering [109, 43, 110, 40] or keypoint prediction [36, 111, 112, 35, 37–39, 41]. To date, RAGNet [46] is the only work that extends LISA [14] to affordance-level segmentation. However, it still lacks interaction-centric modeling and the capability to process video data. The most closely related work, RealVLG-R1 [47], employs

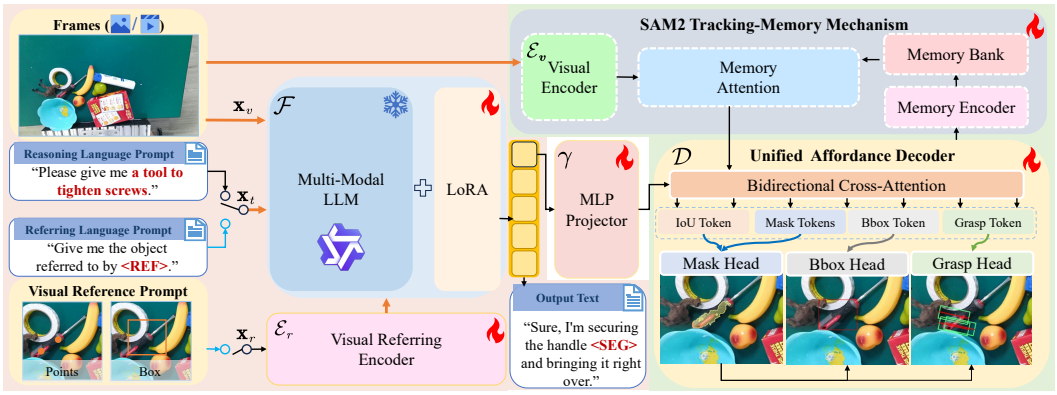


Figure 2: **Overview of UniRAP Model.** Given visual input $\mathbf{x}_v$, text prompt $\mathbf{x}_t$, and optional visual reference prompts $\mathbf{x}_r$, a multimodal model $\mathcal{F}$ produces <SEG>-conditioned outputs. When <REF> is present, visual prompts are encoded as embeddings and jointly modeled with text within a shared <SEG> representation space. Concurrently, visual features from $\mathbf{x}_v$ are aggregated with memory bank representations via memory attention. The resulting features are aligned with <SEG>-derived embeddings through an MLP projector and decoded by an affordance decoder $\mathcal{D}$ to generate segmentation masks, region detections, and 4-DoF interaction poses for unified vision–language spatiotemporal actionable perception.

reinforcement fine-tuning of VLMs to generate structured text for predicting bounding boxes and grasp poses, but is limited by unstable text representations, weak reasoning ability, and lack of video support. In contrast, UniRAP is the first to achieve end-to-end, reasoning-aware unified modeling of actionable affordance perception across both images and videos.

3 UniRAP

Overview. To enable a unified model for vision–language spatiotemporal actionable perception, UniRAP is built upon Qwen2.5-VL [8] and SAM2 [113]. As illustrated in Fig. 2, given a visual input (image or video) $\mathbf{x}_v$, a textual prompt $\mathbf{x}_t$, and optional visual reference prompts $\mathbf{x}_r$ (e.g., points or bounding boxes), UniRAP first leverages a multimodal large model $\mathcal{F}$ to generate textual responses containing <SEG> tokens. When the textual prompt $\mathbf{x}_t$ includes a <REF> token, the visual prompts are encoded into discrete embeddings via a visual referring encoder and injected into the textual sequence, which is then jointly processed by the multimodal model $\mathcal{F}$. These two forms of prompting are unified within a shared representation space centered on <SEG> tokens, facilitating transferability and generalization to tasks without explicit language referring cues. In the visual branch, the input $\mathbf{x}_v$ is encoded by a visual encoder $\mathcal{E}_v$, and the resulting features are fused with historical frame representations stored in a memory bank via a memory attention mechanism, enhancing temporal consistency and object persistence modeling. The fused features are then aligned with implicit prompt embeddings projected from <SEG> tokens through a MLP projector γ, and jointly fed into a unified affordance decoder $\mathcal{D}$. The model yields multi-task predictions, including affordance segmentation masks, region detection results, and interaction poses (4-DoF grasp poses), thereby enabling fine-grained actionable perception in complex spatiotemporal scenarios.

3.1 Unified Vision-Language Spatiotemporal Modeling

To address vision–language spatiotemporal affordance perception under low-data conditions, UniRAP extends the "Embedding as Mask" paradigm [14, 101, 106, 15] by reformulating the problem as a conditional multimodal generation framework based on interface tokens. Specifically, the model takes as input visual observations (images or videos) $\mathbf{x}_v$, textual prompts $\mathbf{x}_t$, and optionally visual reference prompts $\mathbf{x}_r$ (e.g., points or bounding boxes). Through a unified interface token mechanism, it generates reasoning-aware textual outputs, thereby enabling spatial alignment and joint modeling of textual and visual referents. This design maps supervisory signals from heterogeneous modalities into a shared representation space, which not only facilitates cross-task co-optimization but also enhances the model's transferability and generalization performance in multi-task settings.

Specifically, to enable effective transfer learning in scenarios without explicit textual referring expressions, UniRAP introduces a conditional visual referring prompt mechanism. When the input text $\mathbf{x}_t$ contains the <REF> token, the visual prompt $\mathbf{x}_r$ is encoded by a visual referring encoder $\mathcal{E}_r$ into a discrete embedding, which is then injected into the multimodal model $\mathcal{F}$ (Qwen2.5-VL [8]) for joint modeling. The model subsequently generates a reasoning-aware textual output $\hat{\mathbf{y}}$ that includes the <SEG> token. This process can be formally expressed as:

$$\hat{\mathbf{y}} = \mathcal{F}\big(\mathbf{x}_t, \mathbf{x}_v, \mathbf{z}_r\big), \quad \mathbf{z}_r = \begin{cases} \mathcal{E}_r(\mathbf{x}_r), & \text{if <REF>} \in \mathbf{x}_t \\ \mathbf{0}, & \text{otherwise,} \end{cases} \tag{1}$$

where the referring expression encoder $\mathcal{E}_r$ adopts random Fourier mapping [114] to encode the spatial coordinates and temporal information of each point, followed by a linear projection layer. Unlike prior methods [108, 101, 15] designed for visual prompt-based question answering, we adapt this design to unify and align explicit textual referring and visual prompt-based referring within the <SEG> representation space, thereby enabling task transfer with reduced reliance on textual annotations.

Subsequently, to achieve stable and lightweight transfer-based spatiotemporal actionable perception, UniRAP extracts the final-layer hidden representation $\tilde{\mathbf{h}}_{\text{seg}}$ corresponding to the <SEG> token and projects it via a MLP projector $\gamma(\cdot)$ into a unified object-localization prompt embedding:

$$\mathbf{h}_{\text{seg}} = \gamma(\tilde{\mathbf{h}}_{\text{seg}}). \tag{2}$$

In parallel, SAM2 [113] encodes the visual input $\mathbf{x}_v$ into dense features $\mathbf{f}_v$ through its visual encoder $\mathcal{E}_v$. Owing to the tracking-memory mechanism, when processing video sequences, these visual features interact with historical frame features and target states stored in a memory bank, and temporal information is aggregated via memory attention, thereby maintaining temporal consistency across frames. Finally, the memory-enhanced visual features $\mathbf{f}_{\text{mem}}$ and the implicit prompt embedding $\mathbf{h}_{\text{seg}}$ are fused within the decoder $\mathcal{D}$ to jointly predict affordance-related outputs:

$$\hat{\mathbf{M}}, \hat{\mathbf{b}}, \hat{\mathbf{g}} = \mathcal{D}(\mathbf{h}_{\text{seg}}, \mathbf{f}_{\text{mem}}), \tag{3}$$

where $\hat{\mathbf{M}}$ denotes the affordance segmentation mask, $\hat{\mathbf{b}}$ denotes region-level detection results, and $\hat{\mathbf{g}}$ denotes the 4-DoF interaction poses.

3.2 Unified Affordance Decoder

Given that vision–language segmentation data are relatively abundant, whereas annotations for fine-grained interaction perception (e.g., 4-DoF grasp poses) are scarce, we propose a Unified Affordance Decoder (UAD). The goal is to effectively transfer intermediate representations (visual features and implicit prompt embeddings) learned from large-scale coarse-grained perception data to fine-grained tasks, thereby improving affordance perception performance.

Specifically, as illustrated in Fig. 3, the UAD is built upon the Mask Decoder of SAM [113, 24]. During the decoding stage, a Two-way Transformer [63] takes as input the memory-enhanced visual features $\mathbf{f}_{\text{mem}}$, positional encodings $\mathbf{p}$, and task tokens augmented with object-localization prompt embeddings $\mathbf{h}_{\text{seg}}$, including the object pointer token, IoU token, mask tokens $\mathbf{t}_m$, bbox token $\mathbf{t}_b$, and grasp token $\mathbf{t}_g$. Through joint modeling via bidirectional cross-attention and self-attention, the model enables coordinated updates between visual features and token representations, resulting in context-enhanced representations.

To enable unified decoding across different granularities, UAD adopts a token-based feature modulation mechanism for prediction. In the Mask Head, the updated visual features are first upsampled via UpscaleNet to obtain a high-resolution representation $\mathbf{f}_{\text{up}}$. The mask tokens $\mathbf{t}_m$, projected through a HyperNet, are then used to perform element-wise modulation of $\mathbf{f}_{\text{up}}$, producing four candidate mask logits. A stability-based selection strategy is further applied to iteratively refine these candidates, resulting in the final optimal segmentation mask $\hat{\mathbf{M}}$.

For the object detection task, the Bbox Head adopts a DETR-style residual regression modeling strategy [63]. An initial bounding box $\mathbf{b}_0$ is first derived from the optimal segmentation mask $\hat{\mathbf{M}}$ via geometric construction. The bbox token $\mathbf{t}_b$ then modulates the upsampled features $\mathbf{f}_{\text{up}}$ via FiLM [115], followed by mask-guided weighted pooling under $\hat{\mathbf{M}}$. The resulting feature is fed into a ResidualMLP to predict a bounding box offset for refinement. This process can be formulated as:

$$\hat{\mathbf{b}} = \mathbf{b}_0 + \text{MLP}_{\text{res}}(\text{Pool}_{\hat{\mathbf{M}}}(\text{FiLM}(\mathbf{f}_{\text{up}}, \mathbf{t}_b))). \tag{4}$$

To improve robustness, UniRAP further introduces a consistency constraint between the predicted bounding box and the segmentation mask, enabling mutual refinement and suppressing mask noise.

For contact-aware affordance perception, conventional heatmap-based methods [116, 117] exhibit relatively stable predictions but suffer from ground-truth coverage and uneven response distributions. To address these limitations, we reformulate 4-DoF grasp prediction within a regression-based framework, focusing on part-level contact region estimation in the object-centric spatial domain.

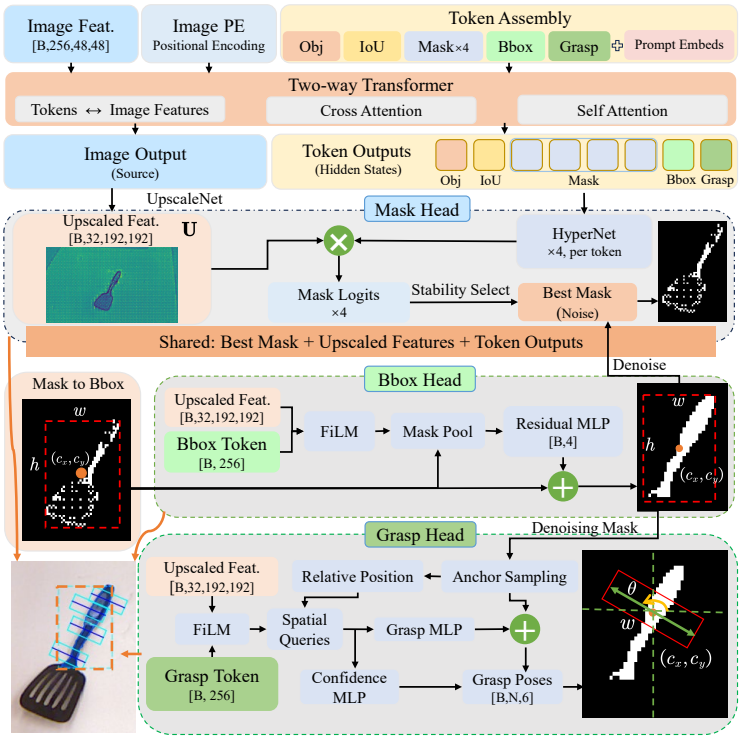


Figure 3: **Architecture of Unified Affordance Decoder.**

Specifically, given a refined segmentation mask $\hat{\mathbf{M}}$, the Grasp Head first applies farthest point sampling (FPS) within the foreground region to select N_g grasp center candidates (a_x^i, a_y^i). Each anchor is encoded as a normalized offset relative to the mask centroid and further scaled by the bounding box dimensions, forming a relative positional encoding $\mathbf{e}_i$ to enhance scale-consistent spatial representation. Next, the upsampled visual feature $\mathbf{f}_{\text{up}}$ is modulated by the grasp token via FiLM conditioning to inject task-specific semantic information. Based on this, N_g learnable spatial queries interact with the modulated features through foreground-masked dot-product attention, producing per-candidate pooled feature representations $\mathbf{z}_i$:

$$\mathbf{z}_i = \text{Attn}\big(\text{FiLM}(\mathbf{f}_{\text{up}}, \mathbf{t}_g),\ \text{FPS}(\hat{\mathbf{M}}),\ \mathbf{e}_i\big). \tag{5}$$

Subsequently, a GraspMLP predicts residual offsets and grasp geometry, while an independent ConfidenceMLP estimates the corresponding grasp quality score q_i:

$$(\Delta c_x^i, \Delta c_y^i, \cos 2\theta_i, \sin 2\theta_i, w_i) = \text{MLP}_{\text{grasp}}(\mathbf{z}_i), \quad q_i = \text{MLP}_{\text{conf}}(\mathbf{z}_i). \tag{6}$$

The final grasp center is obtained via anchor-residual refinement: $c_x^i = a_x^i + \Delta c_x^i, c_y^i = a_y^i + \Delta c_y^i$. We thus obtain N_g grasp candidates, denoted as (see Fig. 3 for illustration):

$$\hat{\mathbf{g}} = \{(c_x^i, c_y^i, \cos 2\theta_i, \sin 2\theta_i, w_i, q_i)\}_{i=1}^{N_g}. \tag{7}$$

Overall, UAD provides a unified decoding paradigm to integrate segmentation, detection, and part-level contact reasoning into a single representation space. By leveraging a shared interface, it enables synergistic knowledge transfer across heterogeneous tasks, ensuring that interaction-centric geometry is grounded in robust semantic features.

3.3 Curriculum Transfer Training

Training Pipeline. To address the imbalance between large-scale segmentation data and limited interaction annotations, we adopt a curriculum-based transfer learning framework that progressively transitions model capabilities from segmentation to interaction-aware perception. As illustrated in Fig. 4, the training pipeline is organized into four stages with increasing task complexity: (1) general

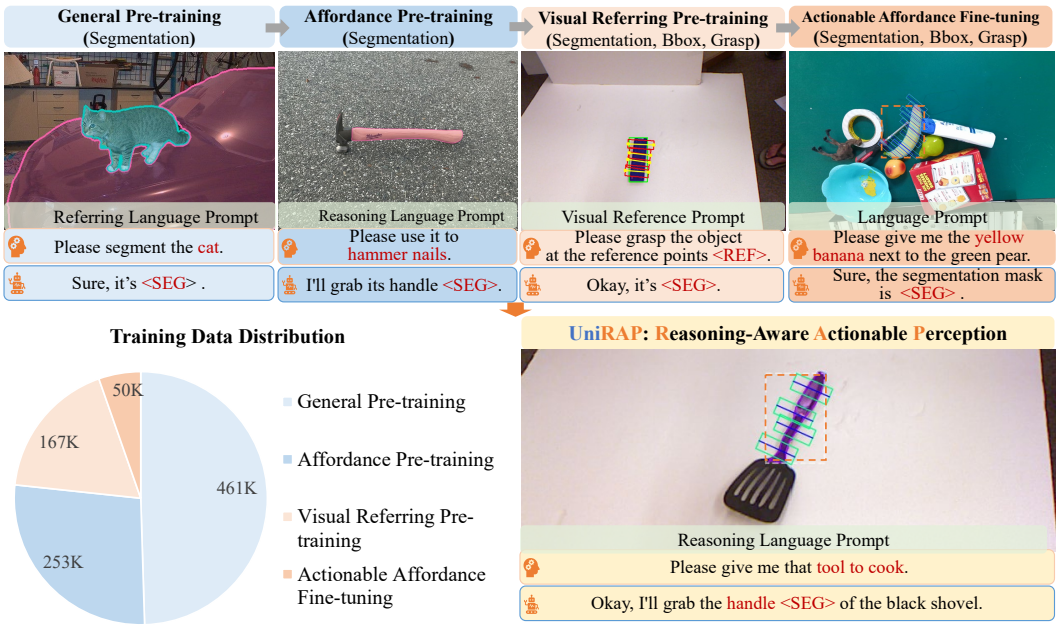


Figure 4: **Curriculum Transfer Training Pipeline.** Details in Appendix B.2.

pre-training with language prompts to enhance visual–semantic alignment and mask prediction; (2) affordance pre-training to localize functionally meaningful regions, bridging category-level segmentation and part-level perception; (3) visual referring pre-training with visual prompts to implicitly learn interaction-relevant poses from grasp datasets without language annotations; and (4) actionable affordance fine-tuning using limited vision-language grasp data to jointly predict segmentation masks, bounding boxes, and 4-DoF interaction poses. This staged strategy enables effective knowledge transfer, thereby improving data efficiency and generalization performance.

Training Objective. The training objective of UniRAP is formulated as a weighted linear combination of a language modeling loss and multiple task-specific losses. The mask decoding component includes a Focal Loss and a Dice Loss for mask prediction, a MAE loss for IoU estimation, and a cross-entropy loss for objectness classification. The object detection branch employs an L1 regression loss and a GIoU loss to supervise bounding box prediction, while the grasp pose estimation branch is optimized using a Huber Loss and a BCE Loss. Furthermore, to enforce spatial consistency across multi-granularity perception, additional constraints are introduced, including a mask-in-bbox loss to ensure containment between masks and bounding boxes, and a consistency loss that aligns grasp center points with the corresponding mask regions. Details are provided in Appendix B.3.

4 Experiments

This section demonstrates that UniRAP preserves strong general segmentation performance while extending to affordance segmentation and actionable perception, enabling a more grounded understanding of functional regions and physical interactions. **Details on implementation (B.1), evaluation metrics (B.4), qualitative results (B.5), and real-world robotic experiments (C.2) are provided in the Appendix.**

4.1 General Segmentation Performance

Referring Expression and Reasoning Segmentation. As shown in Tab. 1, we evaluated UniRAP on RefCOCO/+/g [118, 119] and ReasonSeg [14] to assess its image segmentation capability under both explicit and implicit textual queries. The results demonstrate that UniRAP achieves state-of-the-art performance on referring expression segmentation (RES) benchmarks. Among models of comparable scale, UniRAP significantly outperforms existing methods and, in most settings, attains performance on par with larger models such as UniPixel-7B [15] and Sa2VA-8B [108], indicating a favorable trade-off between parameter efficiency and performance. On ReasonSeg [14], UniRAP remains

Table 1: **Referring Expression and Reasoning Segmentation.** cIoU on RefCOCO/+/g.

Method	Size	RefCOCO [118]			RefCOCO+ [118]			RefCOCOg [119]		ReasonSeg [14]	
		val	testA	testB	val	testA	testB	val(U)	test(U)	gIoU	cIoU
Non-LLM-based Specialists											
ReLA [120]	–	73.8	76.5	70.2	66.0	71.0	57.7	65.0	66.0	–	–
X-Decoder [121]	–	–	–	–	–	–	–	64.6	–	22.6	17.9
SEEM [122]	–	–	–	–	–	–	–	65.7	–	25.5	21.2
LLM-based Image Generalists											
NExT-Chat [102]	7B	74.7	78.9	69.5	65.1	71.9	56.7	67.0	67.0	–	–
PixelLM [101]	7B	73.0	76.5	68.2	66.3	71.7	58.3	69.3	70.5	–	–
LISA [14]	7B	74.9	79.1	72.3	65.1	70.8	58.1	67.9	70.6	61.3	62.9
Groundhog [103]	7B	78.5	79.9	75.7	70.5	75.0	64.9	74.1	74.6	56.2	–
LaSagnA [123]	7B	76.8	78.7	73.8	66.4	70.6	60.1	70.6	71.9	48.8	47.2
M^2SA [105]	13B	74.6	77.6	71.0	64.0	68.1	57.6	69.0	69.3	–	–
LLM-based Video Generalists											
VISA [107]	7B	72.4	75.5	68.1	59.8	64.8	53.1	65.5	66.4	52.7	57.8
Vitron [124]	7B	75.5	79.5	72.2	66.7	72.5	58.0	67.9	68.9	–	–
UniPixel [15]	7B	80.8	83.0	77.4	75.3	80.1	70.0	76.4	77.1	60.5	58.7
Sa2VA [108]	8B	**81.9**	–	–	**76.5**	–	–	**78.9**	–	57.9	44.7
VideoLISA [106]	3.8B	73.8	76.6	68.8	63.4	68.8	56.2	68.3	68.8	**61.4**	**67.1**
Sa2VA [108]	4B	78.9	–	–	71.7	–	–	74.1	–	52.3	44.9
UniRAP (Ours)	3B	**80.9**	**82.6**	**78.1**	**75.1**	**79.3**	**70.4**	**76.1**	**77.5**	57.9	59.0

Table 2: **Referring and Reasoning Video Object Segmentation** on ReVOS `val` [107] split.

Method	Size	Referring			Reasoning			Overall			$\mathcal{R}$
		$\mathcal{J}$	$\mathcal{F}$	$\mathcal{J}\&\mathcal{F}$	$\mathcal{J}$	$\mathcal{F}$	$\mathcal{J}\&\mathcal{F}$	$\mathcal{J}$	$\mathcal{F}$	$\mathcal{J}\&\mathcal{F}$	
Non-LLM-based Specialists											
MTTR [127]	–	29.8	30.2	30.0	20.4	21.5	21.0	25.1	25.9	25.5	5.6
LMPM [128]	–	29.0	39.1	34.1	13.3	24.3	18.8	21.2	31.7	26.4	3.2
ReferFormer [129]	–	31.2	34.3	32.7	21.3	25.6	23.4	26.2	29.9	28.1	8.8
LLM-based Generalists											
LISA [14]	13B	45.2	47.9	46.6	34.3	39.1	36.7	39.8	43.5	41.6	8.6
VISA [107]	13B	55.6	59.1	57.4	42.0	46.7	44.3	48.8	52.9	50.9	14.5
GLUS [130]	7B	56.0	60.7	58.3	48.8	53.9	51.4	52.4	57.3	54.9	17.9
ViLLa [131]	6B	–	–	–	–	–	–	54.9	59.1	57.0	–
UniPixel [15]	7B	**63.9**	**67.8**	**65.8**	**59.4**	**63.7**	**61.5**	**61.7**	**65.7**	**63.7**	**19.4**
Sa2VA [108]	8B	–	–	–	–	–	–	–	–	57.6	–
HyperSeg [125]	3B	56.0	60.9	58.5	50.2	55.8	53.0	53.1	58.4	55.7	–
InstructSeg [126]	3B	54.8	59.2	57.0	49.2	54.7	51.9	52.0	56.9	54.5	–
Sa2VA [108]	4B	–	–	–	–	–	–	–	–	53.2	–
UniRAP (Ours)	3B	**61.8**	**66.5**	**64.1**	**56.8**	**62.4**	**59.6**	**59.3**	**64.4**	**61.9**	**19.4**

Table 3: **Referring Affordance Segmentation.**

Method	Size	HANDAL		GraspNet seen		GraspNet novel		3DOI	
		gIoU	cIoU	gIoU	cIoU	gIoU	cIoU	gIoU	cIoU
Non-LLM Foundation Models									
VLPart [27] + SAM2 [113]	-	40.9	28.9	40.7	27.6	-	-	-	-
G-DINO [58] + SAM2 [113]	-	34.7	26.8	34.9	26.9	-	-	-	-
Florence 2 [132] + SAM2 [113]	-	39.7	22.4	39.4	22.5	-	-	-	-
LLM-based Generalists									
LISA [14]	7B	15.4	11.8	17.7	17.7	25.2	24.1	21.5	13.7
GLaMM [104]	3.8B	25.1	17.0	21.6	10.5	19.2	8.6	19.7	14.1
UniPixel [15]	7B	26.0	16.4	45.0	12.5	48.0	11.5	41.5	24.9
RAGNet [46]	7B	60.5	60.3	63.3	64.0	45.6	33.2	37.4	37.4
UniRAP (Ours)	3B	**63.9**	**65.7**	**70.6**	**64.1**	**51.6**	**42.3**	**52.1**	**37.5**

Table 4: **Reasoning Affordance Segmentation.**

Method	Size	HANDAL (easy)		HANDAL (hard)		3DOI	
		gIoU	cIoU	gIoU	cIoU	gIoU	cIoU
G-DINO [58]	–	3.6	3.0	3.4	3.1	4.1	3.9
LISA [14]	7B	15.5	11.9	12.3	8.1	12.3	8.1
GLaMM [104]	3.8B	4.7	3.5	5.0	3.5	4.4	2.9
UniPixel [15]	7B	26.7	15.7	24.5	11.6	38.5	21.0
RAGNet [46]	7B	58.3	58.1	58.2	57.8	38.1	39.4
UniRAP (Ours)	3B	**63.2**	**64.1**	**62.9**	**63.8**	**55.7**	**54.5**

competitive but does not achieve the best performance. This is primarily because ReasonSeg requires stronger multi-step reasoning capabilities, while the current state-of-the-art method, VideoLISA [106], is specifically designed and trained for such reasoning-intensive segmentation tasks.

Referring and Reasoning Video Object Segmentation. To evaluate UniRAP's ability in mask prediction under implicit textual queries and spatiotemporal reasoning, we conducted experiments on ReVOS [107]. As shown in Tab. 2, UniRAP achieves strong overall performance on ReVOS across both referring and reasoning settings, outperforming representative 3B-scale baselines such as HyperSeg [125] and InstructSeg [126]. It remains competitive compared to recent larger-scale models (e.g., UniPixel-7B), demonstrating a favorable efficiency–performance trade-off.

Table 5: **Actionable Perception Performance on RealVLG-Benchmark** [47].

Method	Size	Seen			Similar			Novel		
		Bbox gIoU/cIoU	Seg. F_β/S_α	Grasp mIoU/gAcc	Bbox gIoU/cIoU	Seg. F_β/S_α	Grasp mIoU/gAcc	Bbox gIoU/cIoU	Seg. F_β/S_α	Grasp mIoU/gAcc
Qwen-VL-Max [135]	-	56.6/57.1	62.8/32.0	2.3/2.4	57.5/57.9	57.6/28.9	2.7/1.3	59.8/60.1	30.3/24.7	1.8/1.2
Gemini2.5-Flash [136]	-	50.1/51.3	2.8/11.8	3.2/1.2	47.9/49.0	3.1/10.6	2.5/2.2	59.5/60.6	3.7/14.2	3.4/1.9
Qwen2.5VL [8]	7B	56.8/58.3	80.6/43.8	1.8/0.8	57.1/57.7	77.7/42.2	1.6/0.0	55.6/56.8	48.3/39.2	1.0/0.0
Qwen2.5VL+SFT [8]	7B	64.8/63.5	82.2/44.1	4.1/2.5	66.7/62.7	79.6/42.6	3.9/2.3	64.9/65.2	48.5/39.4	2.7/1.1
RealVLG-R1(GRPO) [47]	7B	88.0/88.5	89.2/45.3	32.1/39.1	83.4/84.2	85.0/43.8	28.7/36.2	84.6/85.4	52.2/39.7	16.8/18.9
RealVLG-R1(GSPO) [47]	7B	**89.0/89.6**	88.9/**45.4**	33.6/32.8	**86.4/86.9**	86.6/43.8	27.9/30.2	**88.5/88.9**	52.0/39.7	16.5/18.3
Qwen2.5VL [8]	3B	49.6/51.5	74.5/43.2	4.0/2.7	48.4/49.5	71.0/40.1	3.7/1.3	51.3/52.1	44.5/37.5	3.0/1.3
Qwen2.5VL+SFT [8]	3B	56.4/57.1	76.2/43.4	3.4/1.7	54.7/55.9	75.4/43.2	3.6/2.1	57.2/58.6	46.2/38.9	4.4/1.5
RealVLG-R1(GRPO) [47]	3B	83.8/84.2	83.9/43.4	34.7/40.3	84.7/85.3	84.1/43.0	28.2/31.9	77.1/78.1	49.4/38.7	16.3/17.1
RealVLG-R1(GSPO) [47]	3B	87.7/88.4	88.7/45.3	29.2/33.6	84.6/85.3	85.0/43.7	27.6/30.6	78.0/79.3	49.1/39.6	15.3/9.1
UniRAP (Ours)	3B	85.5/86.1	**92.0**/43.6	**43.8/56.7**	83.6/83.0	**89.4/46.9**	**40.8/56.7**	68.4/72.3	**75.8/42.4**	**39.5/63.3**

Table 6: **Ablation Study** of different pre-training stages, fine-tuning strategies, and prediction heads.

Training Strategy				Prediction Head			Evaluation Benchmark				
General Pre-train	Affordance Pre-train	Visual Referring Pre-train	Actionable Affordance Fine-tune	Bbox Head	Heatmap Grasp Head	UniRAP Grasp Head	ReasonSeg (gIoU)	ReVOS (J&F)	3DOI (Ref. gIoU)	3DOI (Reason, gIoU)	RealVLG-Bench (Grasp, gAcc)
✓							55.4	58.3	40.5	37.5	–
✓	✓						56.4	59.2	50.4	52.9	–
✓	✓	✓		✓		✓	57.3	60.2	51.6	54.6	19.6
✓	✓		✓	✓		✓	57.0	59.8	50.8	54.2	26.4
✓	✓	✓	✓			✓	56.8	60.4	51.3	53.8	24.7
✓	✓	✓	✓	✓	✓		57.6	61.7	**52.3**	55.6	13.4
✓	✓	✓	✓	✓		✓	**57.9**	**61.9**	52.1	**55.7**	**63.3**

4.2 Referring and Reasoning Affordance Segmentation Performance

To evaluate UniRAP's capability in affordance segmentation, we followed the evaluation protocol of RAGNet [46] and conducted systematic experiments on the annotated HANDAL [23], GraspNet [133], and 3DOI [134] datasets. Gray indicates zero-shot benchmarks.

As shown in Tab. 3 and Tab. 4, UniRAP achieves strong performance on both explicit text-guided and reasoning-based affordance segmentation tasks. In referring affordance segmentation, UniRAP significantly outperforms RAGNet, which is also designed for affordance modeling, and demonstrates notably better generalization in unseen scenarios. Benefiting from its transfer-learning-oriented design, this advantage becomes even more pronounced in reasoning-based affordance segmentation. On the zero-shot 3DOI dataset, UniRAP improves gIoU and cIoU by 17.6 and 15.1, respectively, indicating stronger robustness and adaptability in complex semantic reasoning and spatial localization.

4.3 Actionable Perception Performance

We evaluated the actionable perception performance of UniRAP on the RealVLG-Benchmark [47], focusing on its capability in both coarse-grained segmentation and fine-grained interaction grounding. Given a natural language description of a target object, the benchmark requires the model to reason about its bounding box, segmentation mask, and 4-DoF grasp pose. Details of the evaluation metrics are provided in Appendix B.4.

As illustrated in Tab. 5, for coarse-grained object detection, RealVLG-R1 [47] retains a certain advantage, benefiting from the large-scale detection pretraining of the foundation model Qwen2.5-VL [8]. However, for segmentation tasks, existing baselines are unable to perform end-to-end prediction and instead rely on a two-stage "bbox + SAM2" pipeline, which introduces error accumulation and leads to slightly inferior performance compared to UniRAP, particularly in zero-shot `Novel` scenarios. For fine-grained 4-DoF grasp pose estimation, text-based generation baselines struggle to capture pixel-level localization, whereas UniRAP's end-to-end formulation demonstrates clear advantages, enabling both stable grasp prediction and stronger generalization. In terms of grasp accuracy (gAcc), especially in novel settings, UniRAP achieves up to a threefold improvement over RealVLG-R1.

4.4 Ablation Study

We conducted ablation studies on selected key components of UniRAP, as summarized in Tab. 6. Affordance pre-training substantially improved affordance understanding and reasoning segmentation, increasing the Referring and Reasoning gIoU on 3DOI by 9.9 and 15.4 points, respectively. Visual referring pre-training further strengthened the association between language instructions and actionable interaction poses, improving grasp accuracy from 26.4 to 63.3 when combined with our

proposed Grasp Head. Actionable affordance fine-tuning jointly optimized segmentation, localization, and grasp prediction, improving grasp accuracy from 19.6 to 63.3 while maintaining competitive segmentation performance. Regarding prediction heads, the Bbox Head provided additional spatial constraints for the SAM2 decoder, suppressing background interference and improving ReasonSeg, ReVOS, and 3DOI performance by 1.1, 1.5, and up to 1.9 points, respectively. Finally, replacing our regression-based Grasp Head with a heatmap-based alternative reduced grasp accuracy from 63.3 to 13.4, demonstrating the effectiveness of direct grasp prediction from Transformer representations.

4.5 Application: Actionable Perception for Robotic Grasping

To evaluate the transferability of UniRAP to real-world robotic manipulation, we conducted grasping experiments using a Franka FR3 robotic arm equipped with an eye-in-hand RealSense D435i camera, as illustrated in Fig. 5. The evaluation includes 13 household objects, with six objects for referring grasping and seven for reasoning-based grasping. We compared UniRAP against RAGNet and RealVLG-R1 under two manipulation paradigms: a modular perception-to-grasping pipeline that combines affordance segmentation with GraspNet, and direct 4-DoF grasp prediction. As shown in Table 7, RAGNet+GraspNet and UniRAP+GraspNet achieve average success rates of 43.8% and 55.4%, respectively, indicating that improved affordance perception alone does not fully resolve the limitations of downstream grasp generation. In particular, the GraspNet-based pipeline remains sensitive to small objects, reflective surfaces, and unreliable depth observations, where inaccurate geometric reconstruction can compromise grasp estimation even when the target region is correctly identified. In contrast, directly predicting actionable 4-DoF grasp poses allows UniRAP to reduce its dependence on a separate geometry-based grasp detector, achieving an average success rate of 88.5%, substantially outperforming RealVLG-R1 (62.3%) and UniRAP+GraspNet (55.4%). These results demonstrate the benefits of unifying language-grounded affordance understanding and grasp prediction for robust real-world robotic manipulation. More details are provided in Appendix C.2.



Figure 5: **Real-World Experimental Setup.** A Franka FR3 robotic arm is equipped with an eye-in-hand RealSense D435i camera. Objects 1–6 evaluate referring grasping, while objects 7–13 evaluate reasoning-based grasping.

Table 7: **Real-world robotic grasping success rates (%).**

Method	Object ID													Avg. SR (%)
	1	2	3	4	5	6	7	8	9	10	11	12	13	
RAGNet + GraspNet	50	60	40	60	50	60	30	40	30	40	30	40	40	43.8
UniRAP + GraspNet	70	70	60	70	60	50	40	70	50	50	40	40	50	55.4
RealVLG-R1 (4-DoF)	80	60	70	80	70	70	50	**80**	40	70	40	20	80	62.3
UniRAP (4-DoF)	**100**	**90**	**80**	**100**	**90**	**100**	**80**	**80**	**70**	**90**	**80**	**90**	**100**	**88.5**

5 Conclusion

We proposed UniRAP, a unified vision-language model that bridges high-level vision-language reasoning with fine-grained physical contact perception. By introducing a token-based interface between the VLM backbone and a dense decoder, UniRAP enables end-to-end mapping from language instructions to interaction-centric geometric representations, supporting affordance segmentation, target region detection, and 4-DoF interaction pose estimation. A curriculum-based transfer learning strategy further improves generalization across heterogeneous tasks. Extensive experiments show consistent gains in both perception and interaction alignment, which establishes a more robust foundation for language-guided robotic systems in dynamic environments. To the best of our knowledge, UniRAP is the first end-to-end, reasoning-aware unified model for actionable affordance perception across both images and videos.

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Appendix

A. UniRAP Model

A.1. Architecture

A.2. Visual Referring Encoder

B. Experiments

B.1. Implementation Details

B.2. Training Recipe

B.3. Training Objective

B.4. Evaluation Metrics

B.5. Qualitative Results

C. Transferring Actionable Perception to Real-World Robotic Manipulation

C.1. Motivation

C.2. UniRAP for Reasoning-Aware Robotic Grasping

D. Discussions

D.1. Limitations & Future Work

D.2. Potential Societal Impacts

E. Licenses

A UniRAP Model

A.1 Architecture

Within the unified vision–language segmentation paradigm, recent approaches [15, 108, 46, 14] commonly adopt an embedding-as-mask design. In this framework, multimodal large language models generate specific tokens (e.g., `<SEG>` [14], `<AFF>` [46], and `TRK` [106]) during text decoding, and the corresponding high-level semantic representations are treated as prompt embeddings to guide segmentation models, such as SAM [24], in producing target regions. This paradigm has been systematically formalized in LISA [14] and its subsequent extensions, and has been widely reused across diverse tasks. Although these methods differ in implementation details and input modalities, for instance, UniPixel [15] supports both image and video inputs and introduces finer-grained supervision via the PixelQA task, they largely share an implicit assumption: visual understanding can be modeled as a direct mapping from semantic representations to pixel space, i.e., a semantic-to-dense prediction process. Under this assumption, variations across methods primarily manifest in task formulation and supervision granularity rather than in the underlying capability paradigm.

However, this assumption becomes inadequate in scenarios involving physical interaction. Taking contact point prediction as an example, the objective extends beyond identifying "what the object is" (semantic level) and "where the object is" (spatial level); it further requires reasoning about "which local region of the object should be contacted" and "whether such contact is geometrically and functionally feasible and stable". This problem inherently introduces additional dimensions of reasoning, including local geometric constraints, potential interaction affordances, and task-relevant functional semantics, thereby exceeding the expressive capacity of conventional segmentation formulations.

Motivated by this observation, UniRAP extends the applicability of the unified paradigm. Rather than treating the `<SEG>` token solely as a semantic anchor for region localization, UniRAP leverages its high-level representation to encode potential physical interaction intent, and, through coupling with a segmentation model, enables a cross-level mapping from semantic understanding to contact reasoning. In contrast to existing methods that primarily focus on pixel-wise correspondences, UniRAP addresses a more structured prediction problem: localizing contact regions in continuous space that satisfy both physical and functional constraints. From a capability perspective, UniRAP therefore does not merely apply existing architectures to a new task, but rather generalizes the paradigm from perception-driven dense prediction to interaction-oriented reasoning under physical constraints. This extension suggests that language-guided segmentation frameworks are not only applicable to conventional visual understanding tasks, but also hold potential for advancing toward higher-level embodied intelligence.

A.2 Visual Referring Encoder

For UniRAP, both point and box prompts are uniformly represented as spatiotemporal coordinates (x, y, t), where bounding boxes are discretized into representative points such as corners or uniformly sampled boundary locations. We adopt a Fourier mapping [114] with random frequency bases to embed low-dimensional coordinates into a high-dimensional representation space.

Specifically, we define the positional encoding as:

$$\gamma(\mathbf{c}) = [\sin(\mathbf{B}\mathbf{c}),\ \cos(\mathbf{B}\mathbf{c})]\,, \quad \mathbf{c} \in \mathbb{R}^3 \tag{8}$$

where $\mathbf{c} = (x, y, t)^\top$, and $\mathbf{B} \in \mathbb{R}^{L\times 3}$ is a random Gaussian matrix whose entries are sampled as:

$$\mathbf{B}_{ij} \sim \mathcal{N}(0, \sigma^2) \tag{9}$$

This random Fourier feature embedding allows the model to capture high-frequency spatial and temporal variations in a linearly separable representation space, which is essential for precise localization and interaction-sensitive reasoning.

Next, the encoded prompt embeddings are transformed into discrete tokens via a learnable linear projection and further interact within LLM, enabling the joint modeling of language-driven semantics and geometry-aware visual prompts. While this design is methodologically inspired by the SAM [24], it differs fundamentally in the pathway of information interaction: the output of the Visual Referring Encoder is no longer fused directly with visual encoder features. Instead, it engages in interaction with the LLM, allowing prompt features to be explicitly aggregated into the semantic representation of the `<SEG>` token, thereby achieving cross-modal alignment within a unified semantic space.

Furthermore, this mechanism is distinct from approaches such as SA2VA [108] and UniPixel [15]. These methods typically convert structured prompts, such as points and bounding boxes, into mask representations for region-based visual question answering tasks. In contrast, our proposed approach adopts a token-level interaction paradigm, mapping tasks without explicit textual annotations into a text-guided `<SEG>` token representation space. This unification reduces distributional discrepancies across tasks at the representation level and mitigates the complexity of cross-task transfer learning.

B Experiments

B.1 Implementation Details

UniRAP is built upon Qwen2.5-VL (3B) [8] and SAM 2.1 [113], with the hidden dimensionality of the visual prompt encoder set to 256.

Within the affordance decoder, all ground-truth annotations are uniformly resampled to a fixed resolution of 768×768 via center padding. This design differs from the direct resizing strategy adopted in SAM2. The motivation is that tasks requiring coordinate inference based on object geometry, such as bounding box estimation and grasp pose prediction, are sensitive to geometric distortions, which can be significantly exacerbated by naive resizing and consequently degrade prediction accuracy.

For bounding box representation, we adopt the parameterization (c_x, c_y, w, h). After center padding, all elements are normalized with respect to the scale of 768. For 4-DoF grasp pose prediction, the representation $(c_x, c_y, \cos 2\theta, \sin 2\theta, w)$ is employed, where (c_x, c_y) denotes the grasp center and θ represents the counterclockwise angle relative to the x-axis. To address the periodicity of angular variables, the angle is encoded as $(\cos 2\theta, \sin 2\theta)$. The parameter w corresponds to the width of the grasp rectangle (see Fig. 3 for illustration). The number of candidate grasp poses N_g is 32.

During training, LoRA is applied to fine-tune Qwen2.5-VL, with the rank and scaling factor set to 128 and 256, respectively. For video inputs, 8 frames are randomly sampled per instance. All experiments are conducted on 8 A800 (80GB) GPUs, and the model is trained for a single epoch.

B.2 Training Recipe

The four stages of Curriculum Transfer Training and their corresponding dataset configurations are summarized in Tab. 8.

Table 8: **Distribution of Training Datasets for UniRAP.** ○ denotes random selection between two prompt types per sample; † indicates text prompts annotated by RAGNet [46]. The fourth stage uses the RealVLG dataset [47]. Stage 1: General Pre-training; Stage 2: Affordance Pre-training; Stage 3: Visual Referring Pre-training; Stage 4: Actionable Affordance Fine-tuning.

Stage	Dataset	Inputs					Outputs				#Samples	#Repeat
		Text	Image	Video	Point	Box	Text	Mask	Box	Grasp		
1	RefCOCO [118]	✓	✓				✓	✓			17K	5
	RefCOCO+ [118]	✓	✓				✓	✓			17K	5
	RefCOCOg [119]	✓	✓				✓	✓			22K	5
	RefClef [118]	✓	✓				✓	✓			18K	5
	ReasonSeg [14]	✓	✓				✓	✓			1.6K	10
	ADE20K [137]	✓	✓				✓	✓			20K	3
	COCOStuff [138]	✓	✓				✓	✓			118K	3
	MapillaryVistas [139]	✓	✓				✓	✓			18K	3
	PACO-LVIS [140]	✓	✓				✓	✓			46K	3
	PASCAL-Part [141]	✓	✓				✓	✓			4.4K	3
	Ref-YouTube-VOS [142]	✓		✓			✓	✓			13K	3
	Ref-DAVIS17 [143]	✓		✓			✓	✓			0.6K	10
	Ref-SAV [108]	✓		✓			✓	✓			56K	3
	MeViS [128]	✓		✓			✓	✓			23K	5
	LV-VIS [144]	✓		✓			✓	✓			11K	3
	ViCaS [145]	✓		✓			✓	✓			41K	3
	ReVOS [107]	✓		✓			✓	✓			29K	5
	GroundMoRe [146]	✓		✓			✓	✓			5.6K	3
2	ReasonSeg [14]	✓	✓				✓	✓			1.6K	10
	†HANDAL [23]	✓	✓				✓	✓			211K	3
	†EgoObjects [147]	✓	✓				✓	✓			12K	3
	†OpenX [148]	✓	✓				✓	✓			11K	3
	†RLBench [149]	✓	✓				✓	✓			5.6K	3
	†GraspNet [133]	✓	✓				✓	✓			12K	3
3	RefCOCO [118]	✓	✓		○	○	✓	✓			17K	5
	RefCOCO+ [118]	✓	✓		○	○	✓	✓			17K	5
	RefCOCOg [119]	✓	✓		○	○	✓	✓			22K	5
	RefClef [118]	✓	✓		○	○	✓	✓			18K	5
	Jacquard [150]	✓	✓		○	○	✓	✓	✓	✓	54K	1
	Cornell [151]	✓	✓		○	○	✓	✓	✓	✓	0.9K	10
	OCID-Grasp [152]	✓	✓		○	○	✓	✓	✓	✓	12K	5
	GraspNet [133]	✓	✓		○	○	✓	✓	✓	✓	26K	2
4	Cornell-VLG [151, 47]	✓	✓				✓	✓	✓	✓	0.9K	5
	OCID-VLG [152, 47]	✓	✓				✓	✓	✓	✓	12K	5
	VMRD-VLG [153, 47]	✓	✓				✓	✓	✓	✓	11K	5
	GraspNet-VLG [133, 47]	✓	✓				✓	✓	✓	✓	26K	5

In the first stage, a segmentation dataset comprising 461K samples is used to pre-train the LoRA adapters, the <SEG> token projection layer, and all parameters of SAM2. Within the Unified Affordance Decoder, only the segmentation head (Seg Head) is optimized, while the remaining components are kept frozen or excluded from training.

The second stage follows the same training configuration as the first stage, but employs a 253K-scale affordance segmentation dataset. Notably, due to the substantial redundancy of scenes in the HANDAL [23] dataset, only half of the samples from each scene are randomly selected for training to mitigate potential bias introduced by duplicated data.

In the third stage, both segmentation datasets with bounding box annotations and grasp datasets without language annotations are utilized to train the visual prompt encoder, as well as the bbox and grasp heads. This stage also involves optimizing the shared Two-way Transformer, with the objective of aligning <REF> prompt features to the feature space of the <SEG> token.

In the fourth stage, the model is initialized with the weights obtained from the previous stage, and only the Unified Affordance Decoder is trained. While preserving the original textual reasoning capability of the vision-language model, this stage further fine-tunes the decoder to align with the prompt features encoded in the <SEG> token space.

B.3 Training Objective

We optimize a weighted sum of task-specific loss functions:

$$\mathcal{L} = \mathcal{L}_{\text{txt}} + 100\,\mathcal{L}_{\text{mask}} + 5\,\mathcal{L}_{\text{dice}} + 5\,\mathcal{L}_{\text{iou}} + 5\,\mathcal{L}_{\text{cls}} + 0.5\,\mathcal{L}_{\text{L1}} + 0.5\,\mathcal{L}_{\text{GIoU}} + 2\,\mathcal{L}_{\text{grasp}} + 0.1\,\mathcal{L}_{\text{mib}} + 0.1\,\mathcal{L}_{\text{gcm}}. \tag{10}$$

The weighting coefficients were determined through validation-based tuning, adhering to segmentation-centric practices. This designates $\hat{\mathbf{M}}$ as the primary spatial signal, to which the remaining branches are anchored.

Language Modeling. The loss $\mathcal{L}_{\text{txt}}$ corresponds to the standard autoregressive cross-entropy computed over the predicted output tokens.

Affordance Mask. Among K candidate masks, only the optimal candidate receives gradient updates:

$$k^* = \arg\min_k \left(\lambda_{\text{mask}} \mathcal{L}_{\text{mask}}^{(k)} + \lambda_{\text{dice}} \mathcal{L}_{\text{dice}}^{(k)} \right), \tag{11}$$

while all other candidates employ stop-gradient. The individual loss terms are defined as follows:

$$\mathcal{L}_{\text{mask}} = -\frac{1}{N} \sum_i \alpha_t (1 - p_t)^\gamma \log p_t, \tag{12}$$

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2 \sum_i \sigma(\hat{\mathbf{M}}_i)\, \mathbf{M}_i + 1}{\sum_i \sigma(\hat{\mathbf{M}}_i) + \sum_i \mathbf{M}_i + 1}. \tag{13}$$

An IoU prediction head produces the estimate $\hat{u}$:

$$\mathcal{L}_{\text{iou}} = \left\| \hat{u} - \frac{|\hat{\mathbf{M}}_{>0} \cap \mathbf{M}|}{|\hat{\mathbf{M}}_{>0} \cup \mathbf{M}|} \right\|_1, \tag{14}$$

where the IoU target is computed from thresholded masks and treated as detached. Supervision is applied exclusively to k^*. An objectness focal loss $\mathcal{L}_{\text{cls}}$ gates all mask-related terms.

Detection. Bounding box predictions $\hat{\mathbf{b}}$ are supervised through the following objectives:

$$\mathcal{L}_{\text{L1}} = \|\hat{\mathbf{b}} - \mathbf{b}\|_1, \quad \mathcal{L}_{\text{GIoU}} = 1 - \text{GIoU}(\hat{\mathbf{b}}, \mathbf{b}). \tag{15}$$

Interaction Pose. Each grasp configuration $(c_x, c_y, \cos 2\theta, \sin 2\theta, w, q)$ is matched to the ground truth via a greedy assignment procedure. The resulting loss combines a Huber loss (applied to the center coordinates, width, and angle) with a binary cross-entropy loss for the quality score.

Spatial Consistency. We introduce two asymmetric constraints to promote spatial coherence. The mask-in-bounding-box loss penalizes activations outside the predicted bounding box $\hat{\mathbf{b}}$:

$$\mathcal{L}_{\text{mib}} = \frac{1}{|\Omega_{\text{out}}|} \sum_{(i,j) \in \Omega_{\text{out}}} \log\left(1 + e^{\hat{\mathbf{M}}_{ij}}\right). \tag{16}$$

Conversely, the grasp-center-in-mask loss encourages the predicted grasp centers to lie within $\hat{\mathbf{M}}$:

$$\mathcal{L}_{\text{gcm}} = \frac{1}{|\mathcal{G}_{\text{valid}}|} \sum_k \log\left(1 + e^{-\hat{\mathbf{M}}(\hat{c}_x^k,\, \hat{c}_y^k)}\right), \tag{17}$$

where $\hat{\mathbf{M}}(\cdot, \cdot)$ denotes bilinear interpolation of the predicted mask evaluated at the specified coordinates. Both loss terms are normalized on a per-object basis and accumulated over the decoding steps.

B.4 Evaluation Metrics

For image segmentation tasks (Tab. 1, Tab. 3, and Tab. 4), we use gIoU and cIoU as the primary evaluation metrics. For video segmentation (Tab. 2), we adopt region similarity $\mathcal{J}$, contour accuracy $\mathcal{F}$, and $\mathcal{J}\&\mathcal{F}$ for performance evaluation. As for the evaluation of hallucination, we adopt the robustness score $\mathcal{R}$ introduced in R2VOS [154].

In the RealVLG-Benchmark (Tab. 5), we follow the evaluation protocol of RealVLG-R1 [47]: object detection is assessed using gIoU and cIoU, segmentation quality is measured by F-measure (F_β) and S-measure (S_α), and grasping performance is evaluated using mIoU and gAcc. Here, gAcc (Grasp Accuracy) is defined as a successful grasp when the predicted grasp box achieves an IoU greater than 0.25 and an angular deviation less than 30°.

B.5 Qualitative Results

Fig. 7 presents qualitative results for Referring Expression and Reasoning Segmentation on RefCOCO/+/g [118, 119] and ReasonSeg [14].

Fig. 8 shows qualitative results for Affordance Segmentation on 3DOI [134], HANDAL [23], and GraspNet `Novel` [133].

Fig. 9 illustrates qualitative results for Referring and Reasoning Video Object Segmentation on ReVOS [107].

Fig. 10 presents qualitative results of UniRAP's actionable perception on the RealVLG-Benchmark [47].

To further evaluate spatio-temporal consistency, we construct pseudo video clips by randomly sampling 6 frames from each scene in the RealVLG-GraspNet subset and arranging them into short sequences. As shown in Fig. 11, UniRAP maintains consistent actionable predictions across frames under variations in viewpoint, occlusion, and scene configuration.

C Transferring Actionable Perception to Robotic Manipulation

C.1 Motivation

Recent Vision–Language–Action (VLA) systems have achieved substantial progress in robotic manipulation tasks. However, these approaches still exhibit notable limitations in fine-grained visual localization required for precise interaction, as well as in their ability to generalize across complex scenes, object variations, and diverse language instructions [29, 30]. In particular, establishing pixel-level correspondences between natural language commands and actionable regions (e.g., grasp points or interaction surfaces) remains challenging. Existing methods often demonstrate instability and insufficient reliability, especially in cluttered or contact-dense environments.

Although perception–action decoupling strategies may not achieve the theoretical upper bound performance of end-to-end approaches, they offer a key advantage in leveraging the generalization capability of perception models. This enables more robust transfer across variations in environments, embodiments, camera configurations, and linguistic instructions. Drawing inspiration from the development trajectory of autonomous driving systems such as Tesla FSD, we argue that a more feasible short-term pathway is to first deploy decoupled solutions in relatively simple scenarios, progressively accumulate embodiment-specific data, and subsequently transition toward end-to-end training in a staged manner.

Accordingly, this work focuses on modeling executable vision–language localization as an intermediate representation for robotic manipulation. Specifically, rather than directly learning an end-to-end control policy, we generate reliable pixel-level or pose-level interaction cues and integrate them with classical geometric planning modules. This hybrid formulation enhances both the robustness and interpretability of robotic execution in complex environments.

C.2 UniRAP for Reasoning-Aware Robotic Grasping

UniRAP can be integrated into robotic grasping pipelines from two complementary perspectives. First, its coarse-grained perceptual outputs, such as bounding boxes or segmentation masks, can be directly combined with geometry-based grasp synthesis methods (e.g., GraspNet [133]). In this setting, UniRAP serves as a semantic front-end that filters task-relevant regions, while downstream geometric planners operate within these regions to generate feasible grasp candidates. This hybrid design preserves the robustness and efficiency of classical pipelines while introducing task-level semantic awareness.

Second, UniRAP produces interaction-aware predictions in the form of 4-DoF grasp poses, which encode both spatial location and orientation constraints aligned with the inferred contact intent. These poses can be directly mapped into the robot coordinate system via standard hand–eye calibration procedures, enabling execution without requiring additional policy learning. Compared to conventional grasp detection outputs that are often agnostic to task semantics, UniRAP's predictions are



Figure 6: **Real-World Experimental Setup.** A Franka FR3 robotic arm is equipped with an eye-in-hand RealSense D435i camera mounted on the end-effector. The test set consists of 13 common household objects, where objects 1–6 are used to evaluate referring grasping performance, and objects 7–13 are used to assess reasoning-based grasping performance.

Table 9: **Object Prompts for Real-World Robotic Vision-Language Grasping.**

Object ID	Object Name	Prompt
		Referring
1	Blue Spoon	Please give me the blue spoon.
2	Orange Cup	Please grasp the orange cup.
3	Razor	Please pick up the grey and black razor.
4	Screwdriver	Please give me the screwdriver.
5	Milk Box	Please give me the milk box.
6	Green Brush	Please grasp the green brush.
		Reasoning
7	Pressure Cooker	I want to braise pork ribs. Can you find a suitable kitchen utensil for me in the image?
8	Blue Stapler	Please give me the blue tool on the induction cooker.
9	Shovel	Please use the tool to stir-fry the vegetables in the pan.
10	Knife	Please use the tool on the table to cut some fruit for me.
11	Rolling Sticker	Please use the tool on the table that can roll and stick lint to help me deal with my clothes.
12	Teapot	I want to make a cup of tea. Please help me find a container that can hold hot water and steep tea leaves.
13	Marker Pen	Please find a pen that can write on a whiteboard and be erased.

Table 10: **Grasp Success Rates (%) on 13 Household Objects.**

Task	Referring						Reasoning							Avg.
ObjectID	1	2	3	4	5	6	7	8	9	10	11	12	13	–
RAGNet [46] + GraspNet [133]	50	60	40	60	50	60	30	40	30	40	30	40	40	43.8
UniRAP + GraspNet [133]	70	70	60	70	60	50	40	70	50	50	40	40	50	55.4
RealVLG-R1 [47]	80	60	70	80	70	70	50	80	40	70	40	20	80	62.3
UniRAP (Ours)	**100**	**90**	**80**	**100**	**90**	**100**	**80**	**80**	**70**	**90**	**80**	**90**	**100**	**88.5**

conditioned on both visual context and language-driven intent, thereby reducing ambiguity in grasp selection.

To evaluate the effectiveness of the contact poses predicted by UniRAP in real-world vision–language robotic grasping tasks, we adopt a pose-based execution protocol. Specifically, the predicted 4-DoF grasp poses are transformed into the robot base frame using hand–eye calibration parameters and subsequently executed directly by the robotic manipulator. As illustrated in Fig. 6, the experimental platform is built upon a Franka FR3 robotic arm equipped with a RealSense D435i eye-in-hand camera mounted on the end-effector. The test set consists of 13 common household objects across different scenarios, with their names and corresponding prompts used in the experiments listed in Tab. 9.

As shown in Tab. 10, we further evaluate the performance of UniRAP and the baseline method RealVLG-R1 (3B) [47] on two types of tasks, namely referring and reasoning language instructions, across 13 test objects, with 10 independent trials per object. RealVLG-R1 directly predicts two contact points to generate grasp poses. Experimental results show that UniRAP achieves a 26.2% higher overall grasp success rate compared to RealVLG-R1, and its performance advantage is particularly pronounced when handling reasoning-based prompts.

In addition, qualitative results are presented in Fig. 12 (Referring) and Fig. 13 (Reasoning), further demonstrating the robustness and generalization capability of UniRAP across diverse scenarios. **Video demonstrations of the experiments are provided at** https://lif314.github.io/projects/unirap**.**

D Discussions

D.1 Limitations & Future Work

Due to computational constraints, we were unable to further scale the training data or increase the model size, which in turn limited potential performance gains. Although incorporating pixel-level visual question answering tasks, such as PixelQA [15], may offer additional benefits, these were not included within the scope of this study.

Given that this work focuses on perception models for embodied scenarios, we argue that it is important to investigate the capability boundaries of smaller models, with an emphasis on achieving real-time actionable perception. Such an approach facilitates deployment on edge devices and reduces the overall computational demands of real-world systems.

D.2 Potential Societal Impacts

This work presents a pixel-level vision–language perception framework tailored for embodied scenarios, enabling a unified formulation that spans coarse-grained object detection and segmentation, functional segmentation, and physical interaction pose estimation. The proposed approach is particularly relevant to application domains requiring frequent and adaptive interaction between agents and their environments, such as robotic home assistance, security, and healthcare. These settings typically demand fine-grained understanding of images and videos, as well as the capability to reason about physical interactions. Moreover, in other embodied applications involving multimodal intelligent agents, the proposed framework can serve as a more expressive and practical alternative.

To the best of our knowledge, this work does not raise any specific potential negative societal impacts that require explicit disclosure.

E Licenses

Our UniRAP is built based on the pre-trained Qwen2.5-VL [8] and SAM2 [113] models. They are both licensed under the Apache License 2.0 (https://www.apache.org/licenses/LICENSE-2.0).

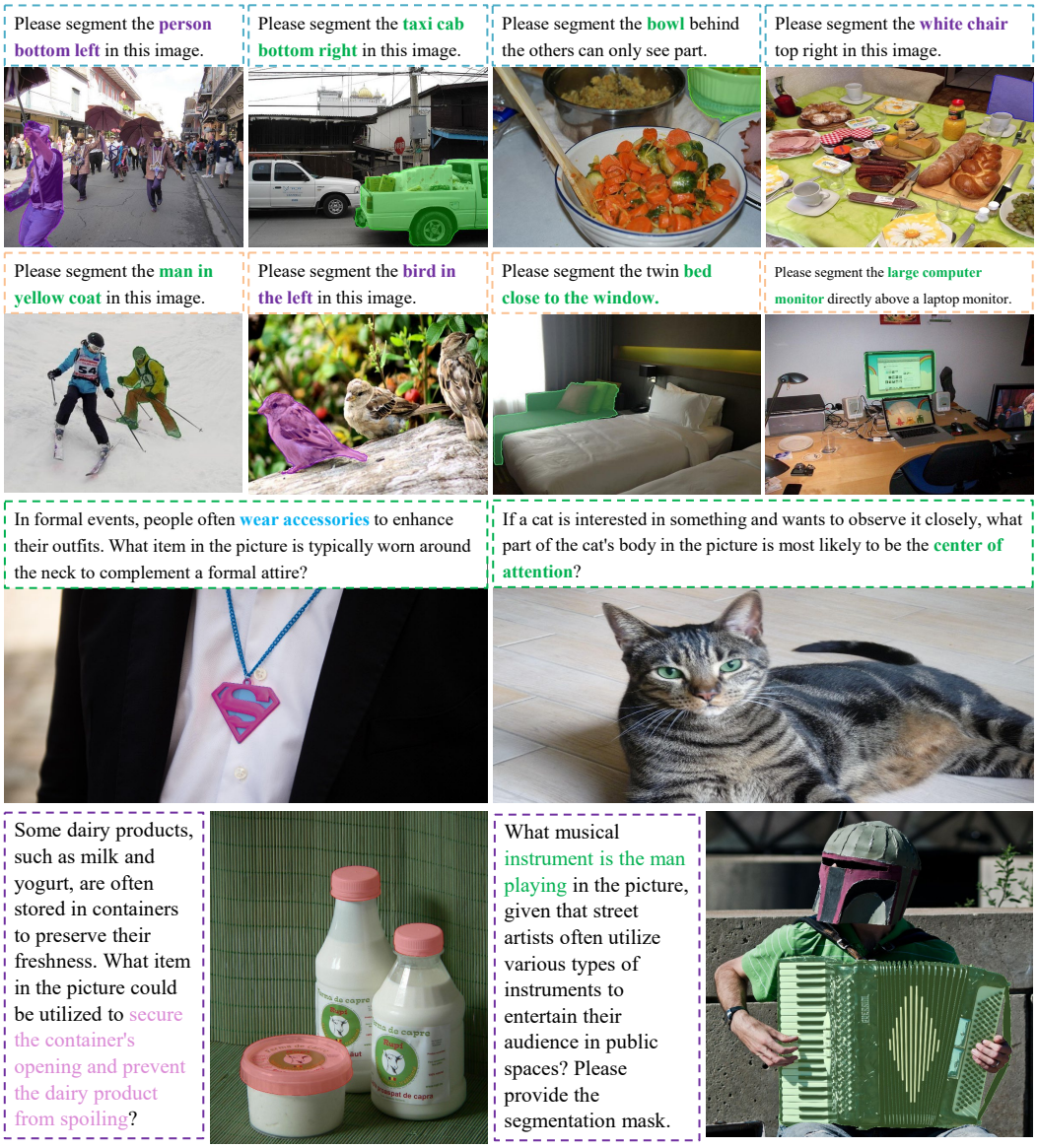


Figure 7: **Qualitative Results of Referring Expression and Reasoning Segmentation.**

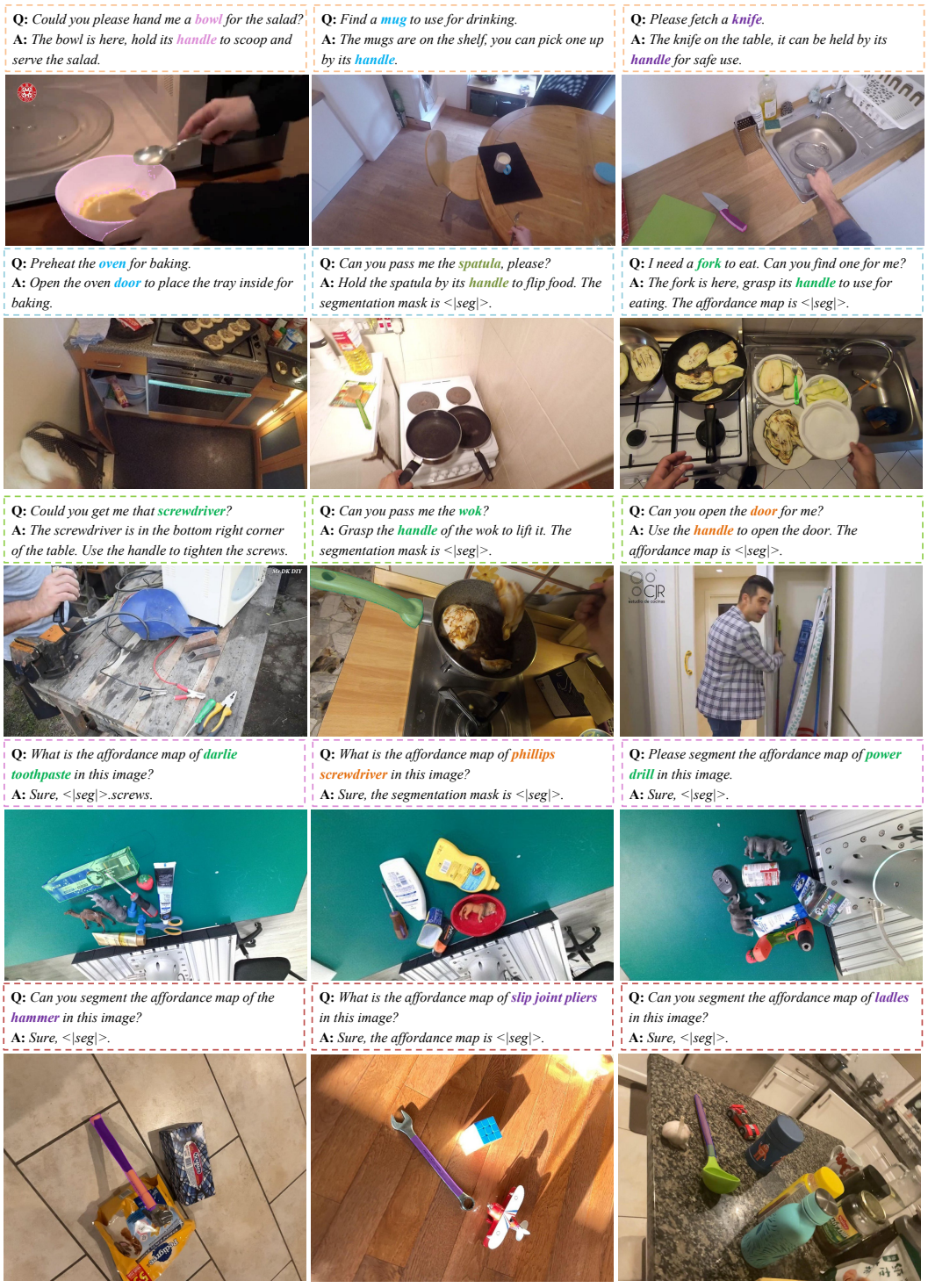


Figure 8: **Qualitative Results of Affordance Segmentation.**

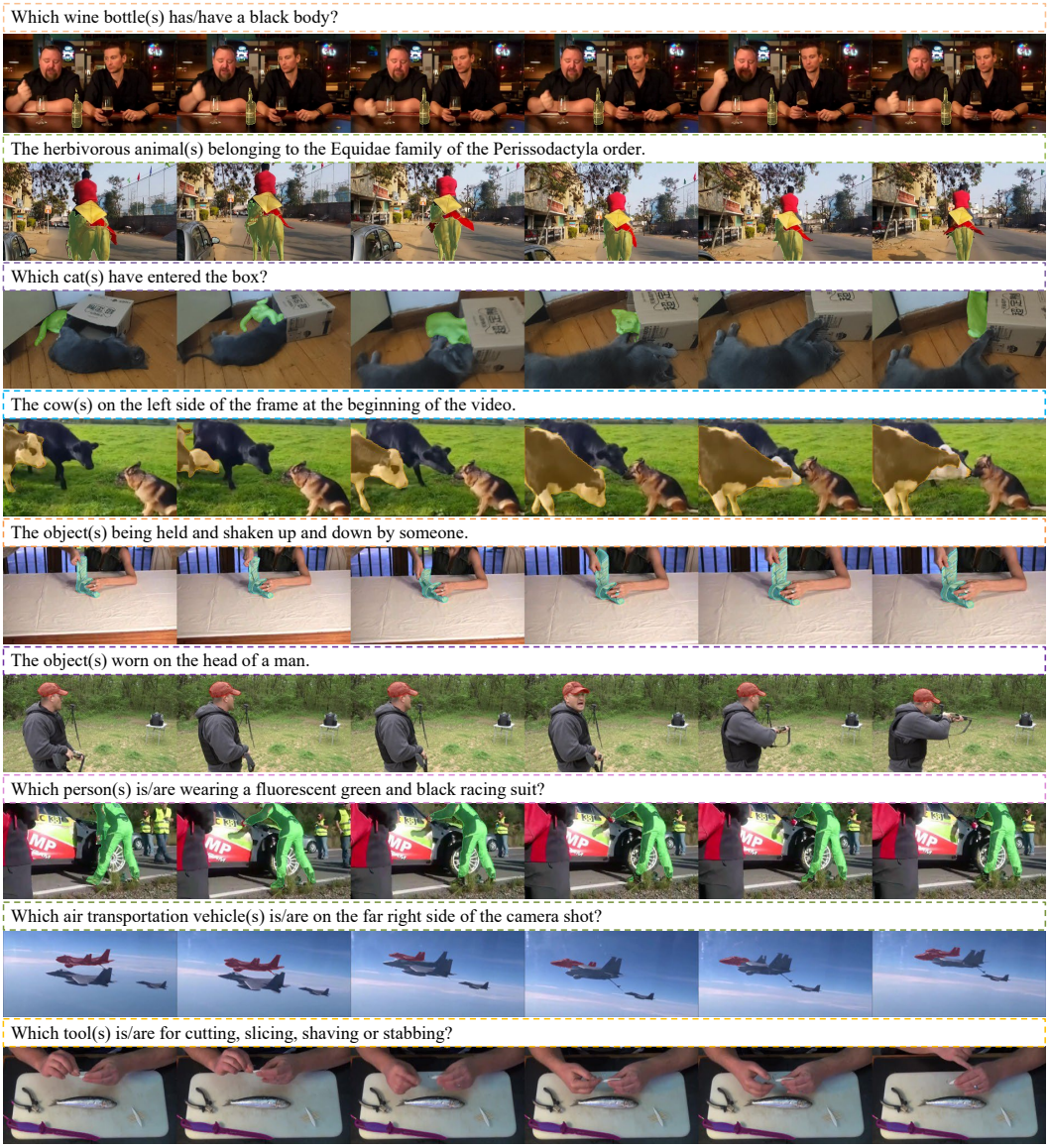


Figure 9: **Qualitative Results of Referring and Reasoning Video Object Segmentation.**

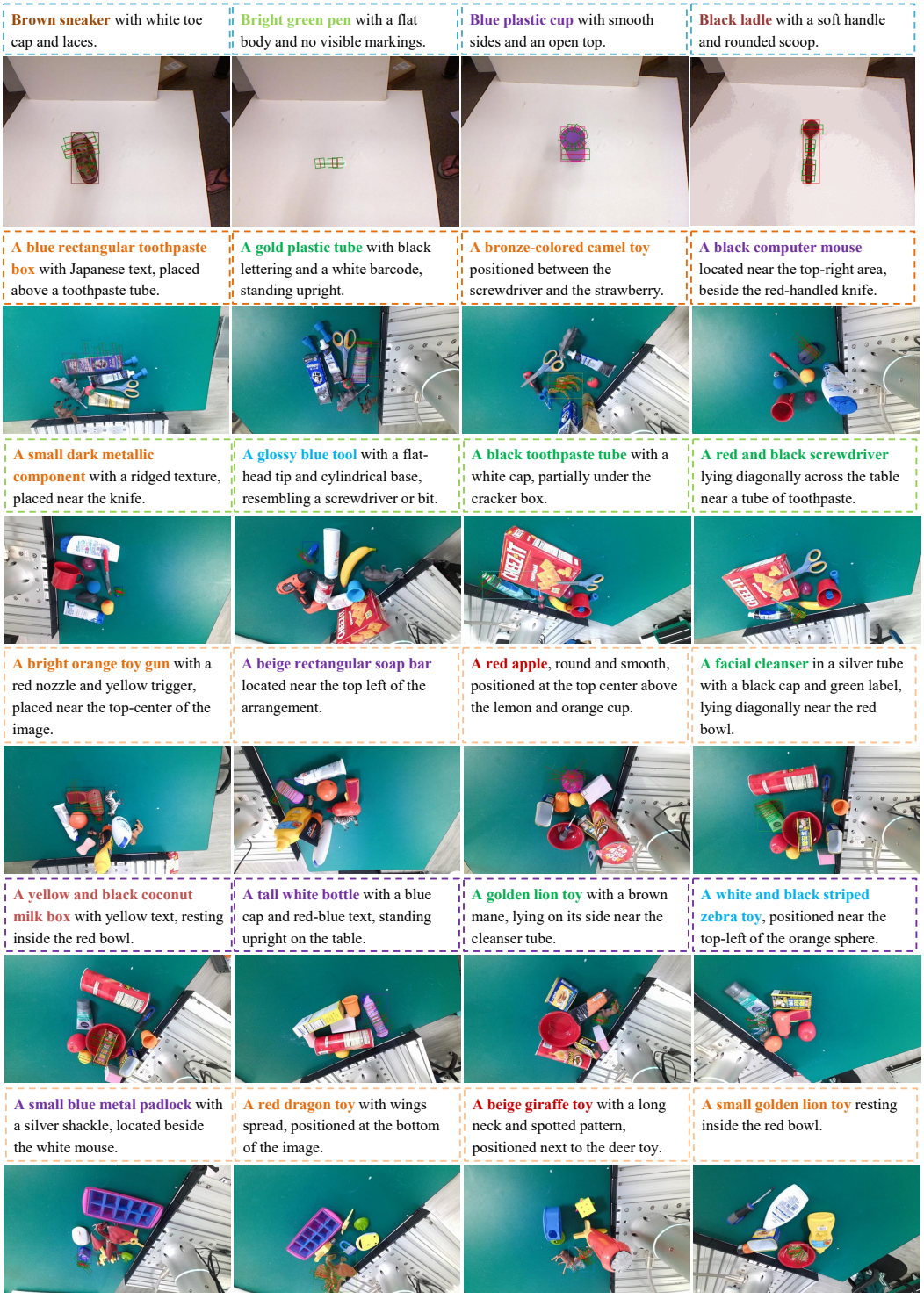


Figure 10: **Qualitative Results of Actionable Perception on RealVLG-Benchmark** [47].

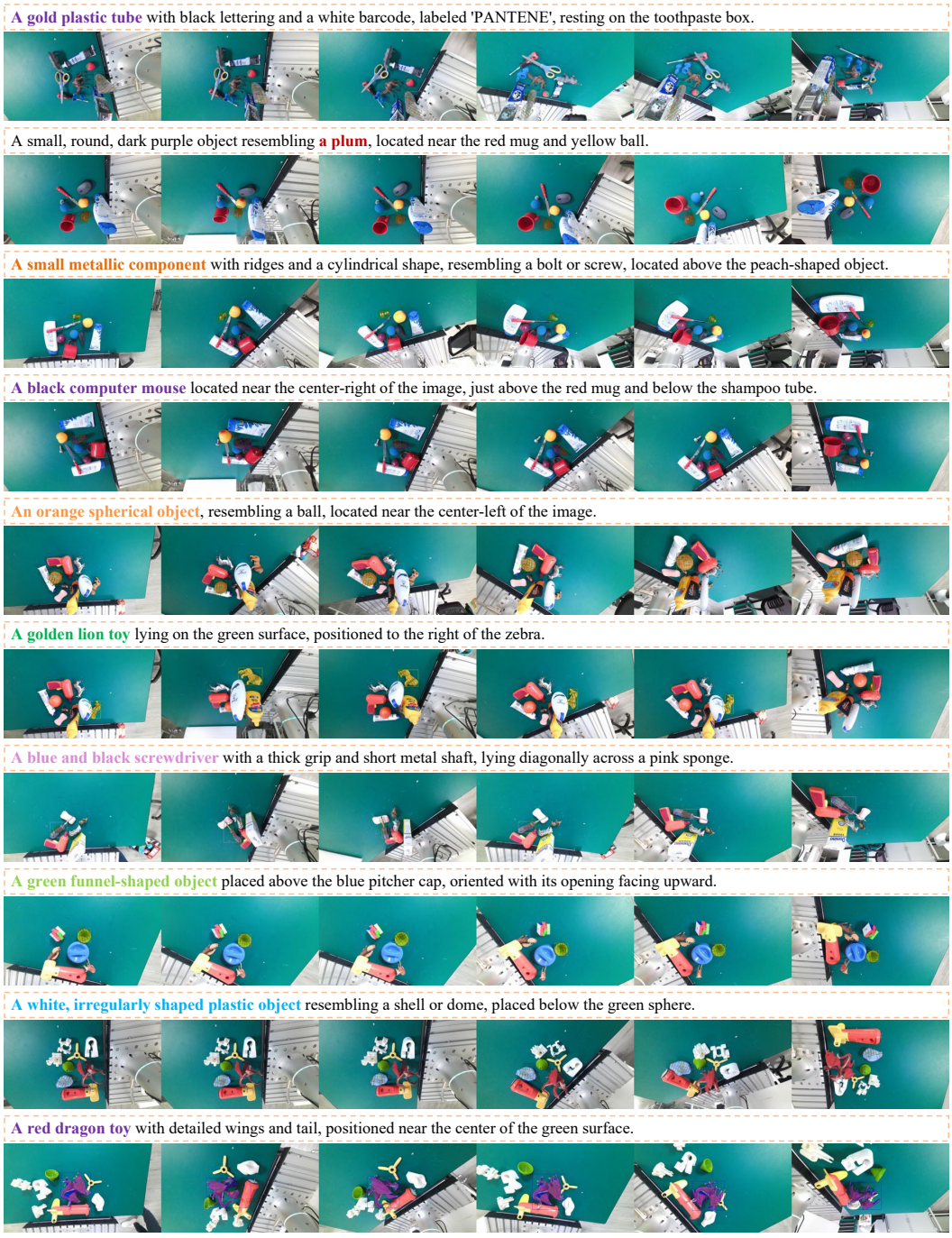


Figure 11: **Qualitative Results of Spatio-Temporal Actionable Perception.** To evaluate the temporal consistency of UniRAP's actionable perception, we construct pseudo-sequences by randomly sampling 6 frames from each scene in the RealVLG-GraspNet subset, due to the absence of video-based grasping datasets.

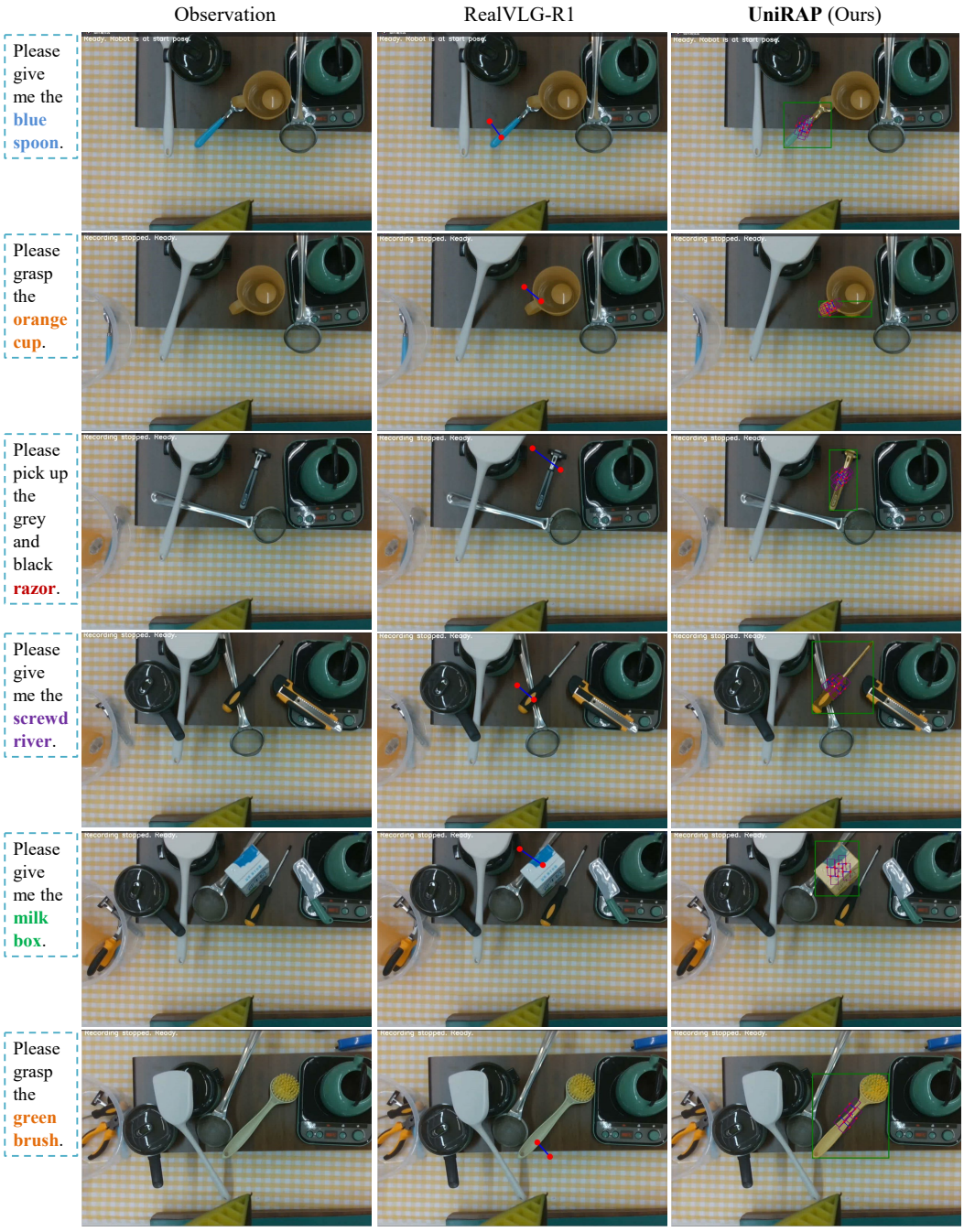


Figure 12: **Qualitative Results of Real-World Referring Vision-Language Grasping.**

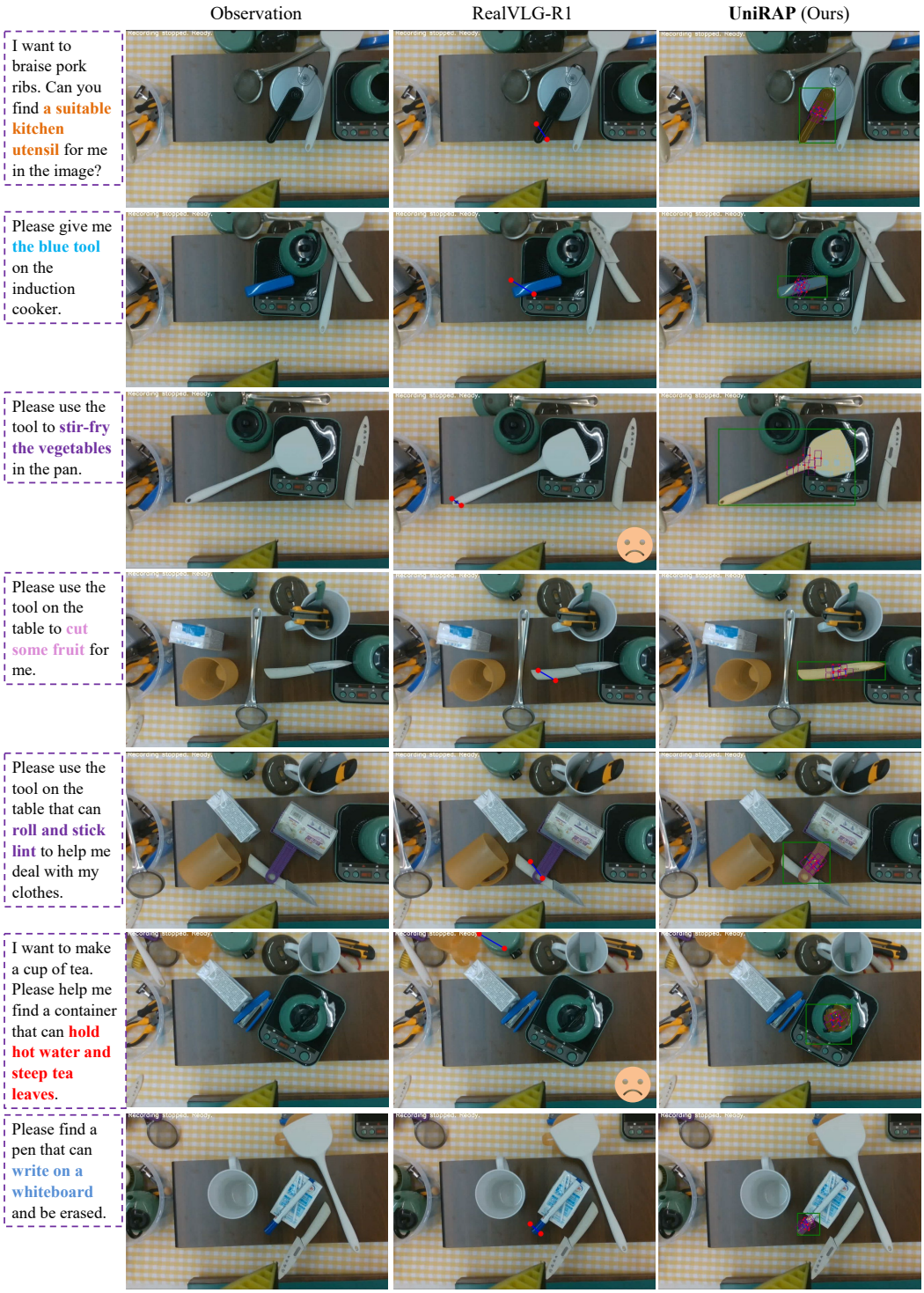


Figure 13: **Qualitative Results of Real-World Reasoning-Based Vision-Language Grasping.** Compared to RealVLG-R1, UniRAP exhibits a stronger affordance modeling capability, enabling it to predict more physically plausible grasp poses, such as correctly identifying appropriate grasp regions on objects like a `Shovel`. In addition, UniRAP demonstrates more robust localization performance. For instance, RealVLG-R1 fails to accurately localize the `Teapot`, whereas UniRAP successfully identifies a valid interaction region and produces a stable grasp prediction.

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